

Session 9: ArraysTopics:1D & 2D Arrays, Array Traversal. Example: Record 7-day screen time & calculate average.

Tasks

- 1) Declare a 1D array called daily Steps with 7 elements to store your step count for each day of the week, assign sample values, and print each value using a loop.**

Code-

```
#include <stdio.h>

int main()
{
    int dailySteps[7] = {5000, 6500, 7000, 8000, 4500, 9000, 10000};

    for(int i = 0; i < 7; i++)
    {
        printf("Day %d steps: %d\n", i + 1, dailySteps[i]);
    }

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc array1.c -o array1.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\array1.exe
Day 1 steps: 5000
Day 2 steps: 6500
Day 3 steps: 7000
Day 4 steps: 8000
Day 5 steps: 4500
Day 6 steps: 9000
Day 7 steps: 10000
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

- 2) Create a 2D array called **playlist Ratings** to store ratings for 3 Spotify playlists over 5 days (rows = playlists, columns = days). Fill it with sample numbers and print the ratings for the second playlist.

Code-

```
#include <stdio.h>
```

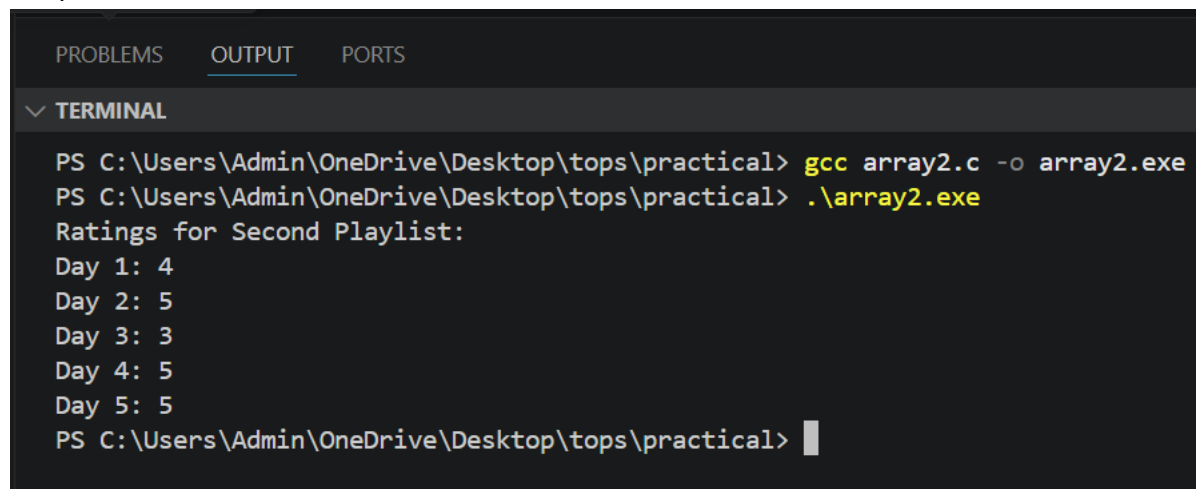
```
int main()
{
    int playlistRatings[3][5] = {
        {5, 4, 5, 3, 4}, // Playlist 1
        {4, 5, 3, 5, 5}, // Playlist 2
        {3, 4, 4, 5, 3}  // Playlist 3
    };

    printf("Ratings for Second Playlist:\n");

    for(int i = 0; i < 5; i++)
    {
        printf("Day %d: %d\n", i + 1, playlistRatings[1][i]);
    }

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc array2.c -o array2.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\array2.exe
Ratings for Second Playlist:
Day 1: 4
Day 2: 5
Day 3: 3
Day 4: 5
Day 5: 5
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> |
```

- 3) Build a function that takes a 1D array of 7 integers representing your daily Zomato order amounts and calculates the average spend for the week.

Hint: Use a loop to sum the values, then divide by the array length.

Code-

```
#include <stdio.h>

float calculateAverageSpend(int orders[], int size)
{
    int sum = 0;

    for(int i = 0; i < size; i++)
    {
        sum = sum + orders[i];
    }

    return (float)sum / size;
}

int main()
{
    int dailyOrders[7] = {250, 300, 150, 400, 350, 200, 500};

    float average = calculateAverageSpend(dailyOrders, 7);

    printf("Average weekly Zomato spend: ₹%.2f\n", average);

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc array3.c -o array3.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\array3.exe
Average weekly Zomato spend: Rs307.14
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

- 4) Given a 2D array called cricket Scores where each row represents an IPL match and columns represent runs scored by each team, write code to print the highest score from each match.

Code-

```
#include <stdio.h>

int main()
{
    int cricketScores[3][2] = {
        {180, 165}, // Match 1: Team 1, Team 2
        {145, 190}, // Match 2
        {210, 205}  // Match 3
    };

    int highest;

    for(int i = 0; i < 3; i++)
    {
        if(cricketScores[i][0] > cricketScores[i][1])
        {
            highest = cricketScores[i][0];
        }
        else
        {
            highest = cricketScores[i][1];
        }
    }
}
```

```
}

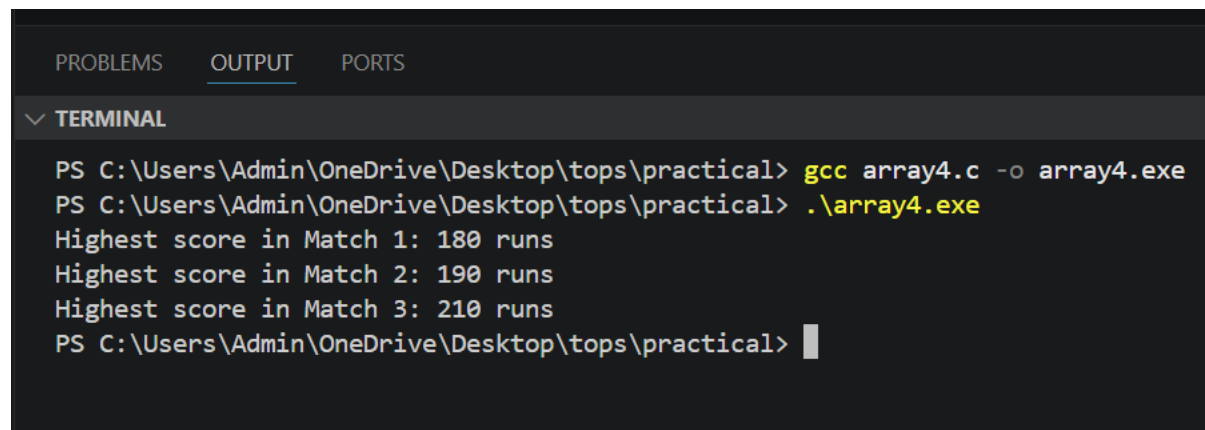
printf("Highest score in Match %d: %d runs\n", i + 1, highest);

}

return 0;

}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc array4.c -o array4.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\array4.exe
Highest score in Match 1: 180 runs
Highest score in Match 2: 190 runs
Highest score in Match 3: 210 runs
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```